Ironbridge National Bank Capstone

Team 4 - Dominic Yeakel and Jason Man



Ironbridge Infrastructure and Host Hardening

AZURE NETWORK
Team 4 subnet 10.10.60.192/26, reached only through Azure Bastion

NSG SEGMENTATION
Bastion in on 22/3389, own team allowed, other teams denied, default deny

IDENTITY
AD DS ironbridge.local with bank OUs and AGDLP security groups

INTERNAL PKI
AD CS Ironbridge Root CA issues TLS for web, RDP, WinRM and LDAPS

Linux web server

`vm-team4-dvwa · 10.10.60.197 · Apache`

- Fully patched; unattended security upgrades on
- UFW default deny; only 22, 80 and 443 open
- SSH: root login off, key-only, named admins
- snapd and unused VMware services disabled
- Web root root:root 755/644; server-status removed
- CA-issued cert, TLS 1.2/1.3 only, HTTP → HTTPS
- auditd rules on identity, SSH, systemd; Fail2ban

Windows domain controller

`vm-team4-dc · 10.10.60.196 · Server 2022`

- Patched; SMBv1, PS v2, Spooler, NetBIOS off
- 14-character passwords, lockout after 5 tries
- Firewall default deny; RDP/WinRM Bastion-only
- RDP and WinRM HTTPS (5986) on CA-issued certs
- Auditing: logons, accounts, privilege use, files
- BankData share: group-only access, synthetic data
- Sysmon; Defender + 13 ASR rules; PS logging; LSA

Incident Response | October 13, 2026

11:03 AM	**DETECTION**	SOC reported HTTP flood against the web server and DC Security log cleared.
12:42–12:50	**INVESTIGATION**	Defacement tickets received; altered website confirmed. Suspicious privileged account ironsvc found, evidence collected and hashed at 12:47 PM
1:01–1:12	**CONTAINMENT**	Apache service stopped; suspected ironsvc account disabled. Attacker revealed to be an employee in the sales department trying to exfiltrate data, but was quarantined by physical security
1:15–1:56	**ERADICATION**	ironsvc removed; suspicious script deleted at 1:26 PM. ApacheBench and flood logs examined; team marked eradication complete.
1:58–2:04	**RECOVERY**	Apache restarted at 1:58 PM; incident closed at 2:04 PM.

Lessons Learned

What we did well

- Dividing roles to each member
- Actively communicating with team members & asking for help
- Staying calm under pressure
- IC supporting other teammates when under duress
- Documentation of the incident
- Declaring each stage of the incident

What we can improve

- Executing commands timely and efficiently
- Utilizing playbooks/runbooks
- Giving updates for role duties at each stage
- Staying organized on the next step

Course Takeaways

Document Throughout, Not at the End

- Documentation should happen continuously during security investigations, incident response, and system changes
- Recording actions, evidence, timestamps, and decisions as they occur creates a reliable audit trail and improves accountability

Humans Are the Weakest Link in Cybersecurity

- Even organizations with advanced security controls can be compromised through human error
- Social engineering attacks, weak passwords, phishing interactions, and improper handling of sensitive information show why employee awareness and secure processes are essential

Proactively plan, effectively execute

- Strong security outcomes start before an incident occurs, with defined response plans, assigned roles, and tested procedures already in place
- Well-rehearsed plans let teams act quickly and confidently under pressure, reducing downtime, limiting damage, and avoiding costly improvised decisions

Summary and Questions

System Hardening

- Secured Ironbridge National Bank’s Azure infrastructure by hardening our Windows Domain Controller and Linux web server through network segmentation, access controls, TLS encryption, and monitoring

Incident Response

- We investigated an insider threat’s attacks including website defacement, HTTP flooding, and audit log clearing; We preserved evidence, disabled the privileged account, and removed a malicious script

Key Takeaways

- Demonstrated how proactive hardening, coordinated incident response, documentation, and thorough recovery validation work together to protect a banking environment

Questions?